MILITARY SPECIFICATION

NETWORK B: 10526468

1. SCOPE

1.1 This specification covers one type of circuit-board, known as Network B. The circuit-board consists of an anode driver, cathode drivers, cable driver, input inverter, and gates used in the Gun Direction Computer: M18.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel: Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction: M18; Manufacturing Processes and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

U.S. Army Munitions Command

F10526468	Network B

FSC 1220

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PACKAGING DATA SHEET

M14002 Data Sheet, Master Packaging

(Copies of specifications, standards, drawings, and packaging data sheets, required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Fabrication.- The network shall be manufactured in accordance with Drawing F10526468 and drawings pertaining thereto.

3.2 General specifications.- The network shall meet the following requirements of MIL-F-13926, and MIL-C-46357.

3.2.1 MIL-F-13926

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Workmanship

3.2.2 MIL-C-46357

a. Etched circuit assembly
b. Marking
c. Handling

3.3 Environmental service conditions.

3.3.1 Vibration.- The network shall be subjected to a simple harmonic motion having an amplitude of 0.015 inch (0.030 inch total excursion). The frequency shall be varied uniformly from 10 to 55 cycles per second (cps) and return to 10 cycles. This cycle shall be traversed in 1 minute ± 15 seconds. The motion shall be applied to the network for a period of 10 minutes in each direction of the 3 mutually perpendicular axes (total of 30 minutes), relative to its normal operational position. Subsequent to vibration, the network shall not be damaged or adversely affected and shall meet the requirements of 3.4.

3.3.2 Shock.- The network shall withstand 3 shock impulses in the vertical plane relative to its normal mounting position. The acceleration of the shock impulses shall be 5 gravity units (G's). Pulse duration of each impulse shall be 11 ± 1 milliseconds. Subsequent to shock the item shall not have been damaged or adversely affected and shall meet the requirements of 3.4.

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3.3.3 Temperatures.

3.3.3.1 Operating.- The network shall meet the requirements of 3.4.1.4, 3.4.2.4, 3.4.5.5, and 3.4.6.5 while exposed to temperatures of -40°F and +135°F.

3.3.3.2 Storage.- The network shall meet the requirements of 3.4 at a temperature of between 60°F and 90°F after having been exposed to each temperature of -80°F and +160°F.

3.3.4 Humidity.

3.3.4.1 Operating.- The network shall meet the requirements of 3.4.1.4, 3.4.2.4, 3.4.5.5, and 3.4.6.5 while exposed to an atmosphere with a relative humidity (RH) up to 97% at a temperature of 135°F.

3.3.4.2 Non-operating.- The network shall meet the requirements of 3.4 after exposure to (and when thoroughly dried) an atmosphere with a RH of 100 percent including condensation, for a period of 4 hours.

3.3.5 Fungus resistance.- The network shall show no evidence of damage and shall meet the requirements of 3.4, after being inoculated with the composite spore suspension specified in Specification MIL-F-13926 and incubated at an ambient temperature of between 77°F and 84°F with a RH of 97 ± 2%.

3.3.6 Salt spray.- The network shall show no evidence of physical damage and meet the requirements of 3.4 immediately after exposure to a 5% salt solution concentration. The solution shall have a specific gravity of from 1.0268 to 0.0413 and a pH between 6.5 and 7.2 at a temperature between 93° and 97°F.

3.4 Performance.- Unless otherwise specified the network shall meet the performance requirements of this specification at ambient temperatures between 60° and 90°F; at a barometric pressure of 28 to 32 inches of mercury; and a RH up to 90%.

3.4.1 Cable driver.

3.4.1.1 Input diode.- The input diode to the cable driver shall have a forward voltage drop of less than 0.5 volt and a reverse leakage of less than 30 microamperes.

3.4.1.2 Saturation voltage.- The input and output transistor, when saturated, shall have a collector to emitter voltage of less than 0.3 volt.

3.4.1.3 Clamp diode.- The clamp diode across the output driver shall have a voltage drop of less than 0.6 volt.

3.4.1.4 Dynamic test.- With an input test signal, the output waveform shall possess the characteristics of that in Figure 1. The delay time measured from the time the input is at minus 1.5 volts (going towards zero volts) to the time the output reached 90 percent of its true level (going towards

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minus 6 volts), shall be less than 0.15 microsecond. The delay time measured from the time the input is at minus 1.5 volts (going towards minus 6 volts) to the time the output reaches 90 percent of its false level (going towards minus 2.3 volts) shall be less than 0.15 microsecond.

3.4.2 Input inverter.

3.4.2.1 Output diode.- The output diode shall have a forward voltage drop of less than 0.8 volt and a reverse leakage of less than 50 microamperes.

3.4.2.2 True input.- The saturation voltage of the transistor shall be less than 0.3 volt and the emitter to base voltage shall be less than 0.6 volt.

3.4.2.3 False input.- The base to emitter diode forward voltage drop shall be less than 0.5 volt. The minus 6 volts clamp diode forward voltage drop shall be less than 0.6 volt.

3.4.2.4 Dynamic test.- With a square wave input signal whose frequency is 5KC ± 10 percent varying from 0 ± 1 volt to minus 6 ± 0.5 volts, the output waveform shall possess the characteristics shown in Figure 2. The delay time measured from the time the input is at minus 3 volts (going towards zero volts) to the time the output reaches minus 3 volts (going towards minus 6 volts) shall be less than 0.5 microsecond. The delay time measured from the time the input is at minus 3 volts (going towards minus 6 volts) to the time when the output reaches minus 3 volts (going towards zero volts) shall be less than 0.5 microsecond.

3.4.3 AND gate.

3.4.3.1 Forward drop.- The diodes shall have a forward drop of less than 0.6 volt dc.

3.4.3.2 Reverse leakage.- The diodes shall have a reverse leakage of less than 50 microamperes.

3.4.4 Clamp circuits and spares.

3.4.4.1 Forward drop.- The diodes shall have a forward drop of less than 0.8 volt dc.

3.4.4.2 Reverse leakage.- The diodes shall have a reverse leakage of less than 50 microamperes.

3.4.5 Anode driver.

3.4.5.1 Reverse leakage.- The reverse leakage of the input diodes shall be less than 50 microamperes.

3.4.5.2 Forward drop.- The forward drop of the input diodes shall be less than 0.8 volt.

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3.4.5.3 True state.- The base to emitter voltage of the input and output transistor shall be less than 1.2 volts dc. The second stage clamp diode forward voltage drop shall be less than 1.0 volt. The output transistor saturation voltage shall be less than 1.0 volt.

3.4.5.4 False state.- The leakage of the input transistor shall be less than 5 microamperes. The second state saturation and base to emitter voltages shall be less than 1.0 volt and 1.2 volts respectively.

3.4.5.5 Dynamic test.- With a square wave input signal whose frequency is 200 cps ± 10 percent varying from 0 ± 1 volt to minus 6 ± 1 volts, the output waveform shall possess the characteristics shown in Figure 4. The delay time measured from the time the input is at minus 1.5 volts (going towards minus 6 volts) to the time the output reaches plus 125 volts (going towards plus 142 volts) shall be less than 100 microseconds. The delay time measured from the time the input is at minus 1.5 volts (going towards zero volts) to the time when the output reaches plus 75 volts (going towards plus 68 volts) shall be less than 100 microseconds.

3.4.6 Cathode driver.

3.4.6.1 Output diode.- The output diode shall have a reverse leakage of less than 100 microamperes.

3.4.6.2 Selection and input diodes.- The selection and input diodes shall have a forward voltage drop of 0.6 ± 0.4 volt dc.

3.4.6.3 True state.- The base to emitter forward voltage drop shall be less than 1.2 volts. The collector to emitter voltage drop shall be less than 1 volt.

3.4.6.4 False state.- The output diode forward voltage drop shall be less than 1 volt.

3.4.6.5 Dynamic test loaded.- With the input test signal as specified in Figure 6 the output waveform shall possess the characteristics shown in Figure 6. The delay time measured from the time the input is at minus 3 volts (going towards minus 6 volts) to the time the output reaches 40 volts, shall be less than 5 microseconds. The delay time measured from the time the input is at minus 3 volts (going towards zero volts) to the time the output reaches plus 60 volts (going towards plus 78 volts) shall be less than 5 microseconds.

3.7 Electrical wiring.- Wiring and connections shall be in accordance with electrical schematic, Drawing F10526471.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services confrom to prescribed requirements.

4.1.1 General provisions.- The component inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Submission of product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examinations and tests.- Except where otherwise specified herein, components and sub-assemblies shall be inspected in accordance with the Inspection provisions for components and sub-assembles" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Tables I, II and III and the test methods and procedures in 4.5. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section herein.

4.2.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection: as specified in MIL-STD-105.

4.2.4 Functional tests.- The requirements and tests in Table I shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60°to 90°F).

TABLE I

(100 percent inspection)

Characteristics		Requirements	Test Procedures
101.	Input diode cable driven (C.D.)	3.4.1.1	4.5.2.1.1
102.	Saturation voltage and clamp diode (C.D.)	3.4.1.2, 3.4.1.3	4.5.2.1.2
103.	Dynamic test (C.D.)	3.4.1.4	4.5.2.1.3
104.	Output diode - Input inverter	3.4.2.1	4.5.2.2.1
105.	True input	3.4.2.2	4.5.2.2.2
106.	False input	3.4.2.3	4.5.2.2.3

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TABLE I (Cond)

Characteristics		Requirements	Test Procedures
107.	Dynamic test	3.4.2.4	4.5.2.2.4
108.	Forward drop -"AND" gate	3.4.3.1	4.5.3.3.1
109.	Reverse leakage -"AND" gate	3.4.3.2	4.5.2.3.2
110.	Forward drop - clamp circuit	3.4.4.1	4.5.2.4.1
111.	Reverse leakage - clamp circuit	3.4.4.2	4.5.2.4.2
112.	Reverse leakage - anode driver	3.4.5.1	4.5.2.5.1
113.	Forward drop	3.4.5.2	4.5.2.5.2
114.	True state	3.4.5.3	4.5.2.5.4
115.	False state	3.4.5.4	4.5.2.5.3
116.	Dynamic test	3.4.5.5	4.5.2.5.5
117.	Output diode - cathode driver	3.4.6.1	4.5.2.6.1
118.	Selection and input diode	3.4.6.2	4.5.2.6.2
119.	True state	3.4.6.3	4.5.2.6.3
120.	False state	3.4.6.4	4.5.2.6.4
121.	Dynamic test	3.4.6.5	4.5.2.6.6
122.	Electrical wiring	3.4.7	

4.3 Standard sample.

4.3.1 General.- One network shall be selected at random by a Government representative as a standard sample from each 100 produced. The sample shall meet the requirements and tests in Table II.

TABLE II

Characteristic		Requirement	Test Procedures
301.	Fabrication	3.1	Applicable drawings-visual
302.	General specifications	3.1	MIL-F-13926 and MIL-C-46357 visual

4.3.2 Environmental.- Three units shall be selected at random by a Government representative as a standard sample from each 50 produced or from each month's production, whichever occurs first. The samples shall have met the requirements and tests in Table I and shall then meet the requirements and tests in Table III.

TABLE III

Characteristic		Requirement	Test Procedures
303.	Vibration	3.3.1	4.5.1.1
304.	Shock	3.3.2	4.5.1.2
305.	Humidity-Operating	3.3.4.1, 3.4.1.4, 3.4.2.4, 3.4.5.5, 3.4.6.5	4.5.2.1.4,4.5.2.2.5 4.5.2.5.6,4.5.2.6.6
306.	Humidity-Non-operating	3.3.4.2	4.5.1.3.1
307.	Salt spray	3.3.6	4.5.1.4

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TABLE III (Cond)

Characteristic	Requirement	Test Procedures
308. Operating temperature	3.3.3.1, 3.4.1.4, 3.4.2.4, 3.4.5.5, 3.4.6.5	3.5.2.1.4, 4.5.2.2.5, 4.5.2.5.6, 4.5.2.6.(
309. Storage temperature	3.3.3.2	4.5.1.5

4.3.3 Failure sample - Should any one item of a standard sample fail to meet the specified test requirements, acceptance of the, represented production lot product will be suspended by the Government ultil necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.2.3).

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicabl requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.4.1.1 Test equipment.- In conjunction with 4.4.1, the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.5:

Item Number	Description
1 Oscilloscope	A dual trace oscilloscope with a rise time of less than 50 millimicroseconds and an input impedance of 1 meg. ohms or greater and a input capacitance of less than 30 micro-microfarad.
2 Voltage measuring device	A + 5 percent dc, voltage measuring device with a 10 millivolt to 300 volt range. The input impedance shall be greater than 10 meg. ohms.
3 Pulse generator	A generator of pulses capable of square waves up to to 500KC with a rise and fall time of less than 100 millimicrosecond and an output impedance of less than 150 ohms.

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Item Number	Description
4 Power supply	Minus 1.5 volts + 2 percent with a current requirement of minus 150 milliamperes. The peak-to-peak ripple and noise shall be less than 100mv. The output impedance shall be less than 2 ohms from dc to 500KC.
5 Power supply	Minus 3 volts ± 2 percent current requirement of minus 50 milliamperes. The peak-to-peak ripple and noise shall be less than 100mv. The output impedance shall be less than 2 ohms from dc to 500KC.
6 Power supply	Minus 6 volt ± 2 percent, current requirement of 200 milliamperes. The peak-to-peak and noise shall be less than 400mv and the output impedance shall be less than 2 ohms from dc to 500KC.
7 Power supply	Minus 18 volts ± 2 percent current requirement of 150 milliamperes. The peak-to-peak ripple and noise shall be less than 600mv and the output impedance shall be less than 2 ohms from dc to 500KC.
8 Power supply	Plus 6 volts ± 2 percent current requirement of 150 milliamperes. The peak-to-peak ripple and noise shall be less than 100mv and the output impednace shall be less than 2 ohms from dc to 500KC.
9 Power supply	Plus 15 volts ± 2 percent current requirement of 50 milliamperes. The peak-to-peak ripple and noise shall be less than 100mv and the output impedance shall be less than 2 ohms from dc to 500KC.
10 Power supply	Plus 155 volts ± 2 percent current requirement of 120 milliamperes. The peak-to-peak ripple and noise shall be less than 400mv and the output impedance shall be less than 10 ohms from dc to 500KC.
11. Power supply	Plus 66 volts minus 12, plus 9 percent current requirement of 120 milliamperes. The peak-to-peak ripple and noise shall be less than 200mv, and the output impedance shall be less than 10 ohms from dc to 500KC.
12 Power supply variable	From zero to one hundred and fifty volts current requirement of 100 milliamperes. The peak-to-peak ripple and noise shall be less than 500mv and the output impedance shall be less than 10 ohms from dc to 500KC.

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Item Number		Description
13	Power supply	Plus 143 volts ahll not vary more than 12.5 ± 1 volt from the plus 155 volt (item 11) supply. The current requirement of 50 milliamperes and a peak-to-peak ripple and noise of less than 400mv. The output impedance shall be less than 10 ohms from dc to 500KC.
14	Power supply	Plus 78 volts shall not vary more than 12.5 ± 1 volts from the plus 66 volt (item 11) supply. The current requirement of 50 milliamperes and a peak-to-peak ripple and noise of less than 200mv. The output impedance shall be less than 10 ohms from dc to 500KC.

4.5 Test methods and procedures.

4.5.1 Environmental.

4.5.1.1 Vibration.- The vibration test shall be made on a device with means provided for measuring and controlling the direction of vibration and adjusting the amplitudes and frequencies to keep them within the specified limits of 3.3.1. The measuring instruments of the testing device shall be capable of measuring the vibration amplitude within plus or minus 5 percent and capable of measuring the frequency within plus or minus 2 percent. The test shall be performed in accordance with the requirements specified in 3.3.1 with the network rigidly mounted in its normal operating position. Subsequent to the vibration test, the unit shall show no damage or defect resulting from the test when examined by visual and tactile means. The unit shall then be inspected as specified in 4.5.1.2 prior to determining compliance with requirements specified under 3.4.

4.5.1.2 Shock.- This test shall be conducted in accordance with Procedure II of the shock test defined in MIL-F-13926 except that the network shall be shocked in the vertically plane only. The impact (shocks) and "G" value shall be in accordance with 3.3.2. A receiving device, representing the connector receptacle that receives the network in the computer, shall be secured to the horizontal surface of the shock tester. Each end of the receiving device shall be clamped such that the female receiving slot will hold the unit in a vertical plane. The unit will then be inserted into the receiving device and shocked. Subsequent to imposing the number of impacts required, a visual examination shall be made for damages or defects to determine compliance with 3.3.2 and then the unit will be inspected as specified in 4.5.1.3 prior to determining compliance with 3.4 as required by 3.3.2.

4.5.1.3 Humidity.

4.5.1.3.1 Non-operating.- This test shall be conducted in accordance with the conditions in 3.3.4.2 and the humidity test defined in Procedure I of MIL-F-13926 except that the 4-1/2 hour period shall be reduced to 2 hours the 6 hour period reduced to 2 hours, and the total test period shall not

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exceed 5.5 hours per cycle. The unit shall then be throughly dried and then subjected to the test in 4.5.1.3.2 prior to meeting the specified requirements of 3.4 as required by 3.3.4.2.

4.5.1.3.2 Operating.- This test shall conducted in accordance with the conditions in 3.3.4.1 and the humidity test defined in Procedure III of MIL-F-13926 except that the humidity shall be maintained at 97 ± 5 percent for the 24 hour period, and the test shall cease at the end of that period (Method 106 of MIL-STD-202 shall be excluded). Throughout the test the unit shall be operated in accordance with 4.5.2.1.4, 4.5.2.2.5, 4.5.2.5.6 and 4.5.2.6.6 to determine compliance with 3.3.4.1.

4.5.1.4 Salt spray.- The network shall be subjected to the conditions in 3.3.6 and the salt spray test defined in MIL-STD-202, Condition B. Immediately after exposure, the unit shall be inspected for any physical damage, then tested in accordance with 4.5.3 and 4.5.3.1 for compliance with 3.3.6.

4.5.1.5 Temperature (operating and storage).- After being subjected to all environmental tests specified above, the network shall be subjected to Test Procedure I specified in MIL-F-13926, except the temperature specified in 3.3.3 shall apply. While at the operating temperatures of 3.3.3.1, the unit shall be tested in accordance with 4.5.2.1.4, 4.5.2.2.5, 4.5.2.5.6, and 4.5.2.6.6 to determine compliance with the requirements of 3.3.3.1. The temperature shall then be returned to standard temperature of (60° to 90°F) and the unit shall be tested in accordance with 4.5.2 to determine compliance with 3.3.3.2.

4.5.1.6 Fungus.- To determine compliance with 3.3.5, one of the first 5 networks produced shall be submitted for testing in accordance with the fungus test defined in MIL-F-13926. After approval, subsequent acceptance for fungus requirements may be by certification.

4.5.2 Performance.- All testing procedures shall be performed under temperature conditions. relative humidity and barometric pressure specified in 3.4 unless otherwise specified.

4.5.2.1 Cable driver.

4.5.2.1.1 Input diodes.

4.5.2.1.1.1 Forward voltage drop. Connect to pin 58B a minus 6 volt supply voltage and ground pin 8B. The voltage measured from ground to pin 7B shall be less positive than plus 0.5 volt as specified in 3.4.1.1.

4.5.2.1.1.2 Reverse leakage.- Remove the minus 6 volts from pin 58B and ground pin 58B. Apply the minus 6 volts to pin 8B. The voltage measured at pin 7B shall be more positive than minus 0.30 volt dc as specified in 3.4.1.1. Remove all test leads from board.

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4.5.2.1.2 Saturation voltage.- To determine compliance with the conditions of 3.4.1.2 and 3.4.1.3, connect a 200 ohm 1/2 watt, ± 5 percent resistor from pin 5B to ground. Connect a 68 ohm 1/2 watt, ± 5 percent, resistor from the minus 6 volt supply in pin 5B. Connect to the indicated pins, the supply voltages specified in Table IV. Ground the positive polarities of all negative supply voltages, and ground the negative polarities of all positive supply voltages.

TABLE IV

To Pins	Connect Supply Voltage	Item Number
3A	-1.5V	4
56A	-3.0V	5
58A	-6.0V	6
58B & 40B	-18V	7
10A	+6V	8
11B	+15V	9
56B	+155V	10
4A	+143Z	13
4B	+66Z	11
38B	+78Z	14
12B	Power Ground	

4.5.2.1.2.1 True input.- Connect pin 8B to the minus 6 volt supply. Measure the voltages between the pins as specified in Table V.

TABLE V

From Pins	To Pins	Voltage Less Positive Than:
3A	7B	+0.70
3A	5A	+0.40
5A	6B	+0.60
12B	5B	+4.0

4.5.2.1.2.2 False input.- Remove the minus 6 volt supply from pin 8B and ground pin 8B. Measure the voltage between the pins as specified in Table VI to assure conformance with 3.4.1.2 and 3.4.1.3. At the end of these tests temove the ground from pin 8B.

TABLE VI

From Pins	To Pins	Voltage Less Positive Than:
6B	5A	+0.65
6B	58A	+0.30
12B	5B	+5.50

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4.5.2.1.3 Dynamic test.- With the load and power supplies connected, as specified in 4.5.2.1.2 apply to pin 8B an input test signal as specified in Figure 1, The output waveform at pin 5B shall be as specified in Figure 1. The delay time measured from the time the input is at minus 1.5 volts (going towards zero volts) to the time when the output reached 90 percent of its true level (going toward minus 6 volts), shall be less than 0.15 microsecond. The delay time measured from the time the input is at minus 1.5 volts (going towards minus 6 volts) to the time when the output reached 90 percent of its false level (going towards minus 2.3 volts) shall be less than 0.15 microsecond for compliance with 3.4.1.4. At the end of these tests, remove the resistor load and power supplies connected in 4.5.2.1.2.

4.5.2.1.4 Circuit output.- The cable driver output at pin 5B shall meet the requirements o3 3.3.3.1 and 3.3.4.1 when tested as outlined in 4.5.2.1.3. At the end of these tests remove the resistor load and power supplies.

4.5.2.2 Input inverter.

4.5.2.2.1 Output diodes.

4.5.2.2.1.1 Reverse leakage.- Connect pin 58A to the minus 6 volt supply and ground pin 10B. The voltage measured from pin 58A to pin 8A shall be more positive than minus 1.2 volts as specified in 3.4.2.1.

4.5.2.2.1.2 Forward voltage drop.- Remove the minus 6 volts from pin 58A. Connect to pin 8A a 10K ohm, 1/4 watt, ± 5 percent resistor from a plus 15 volt supply. The voltage measured from pin 8A to pin 10B shall be more negative than plus 0.6 volt as specified in 3.4.2.1. Remove all test leads from board and connect a 2K ohm, 1/2 watt ± 5 percent, resistor from the minus 18 volt supply to pin 10B. The power supplies ahll remain connected as specified in Table IV.

4.5.2.2.2 True input.- Connect to pin 7A the minus 6 volt supply. Measure the voltages between the pins as specified in Table VII for comformance with 3.4.2.2. At the end of this test remove the minus 6 volt supply from pin 7A.

TABLE VII

From Pins	To Pins	Voltage more Negative Than:
12B	6A	+0.60
12B	8A	+0.30
12B	10B	+1.10

4.5.2.2.3 False input.- To determine compliance with the conditions of 3.4.2.3 utilize the same connections for the power supplies and load specified in 4.5.2.2.1.2. Ground pin 7A and measure the voltages between the pins specified in Table VIII to determine compliance with 3.4.2.3. At the end of this test remove the ground from pin 7A.

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TABLE VIII

From Pins	To Pins	Voltage More Negative Than:
6A	12B	+0.50
58A	8A	+0.60
58A	10B	+1.40

4.5.2.2.4 Dynamic test.- With the power supplies and load remaining connected as specified in 4.5.2.2.1.3, apply to pin 7A a 5KC ± 10 percent square wave varying from 0 ± 1 volt to minus 6 ± 0.5 volts as shown in Figure 2. The output waveform at pin 10B shall possess the characteristics shown in Figure 2. The delay time measured from the time the input is at minus 3 volts (going towards zero volts) to the time when the output reaches minus 3 volts (going towards minus 6 volts) shall be less than 0.5 microsecond. The delay time measured from the time the input is at minus 3 volts (going towards minus 6 volts) to the time when the output reaches minus 3 volts (going towards zero volts) shall be less than 0.5 microsecond for compliance with 3.4.2.4.

4.5.2.2.5 Circuit output.- The input inverter output at pin 10B shall meet the requirements of 3.3.3.1, 3.3.4.1, when tested as outlined in 4.5.2.2.4. At the end of these tests remove all test leads and the load specified in 4.5.2.2.1.

4.5.2.3 AND gate test.

4.5.2.3.1 Forward drop.- With the power supplies connected as shown in Table IV, the voltage measured from pin 58A to pins 20B and 15B shall be plus 0.5 ± 0.4 volt dc. Ground the indicated pin as listed in column 1, and as each pin in column 1 is grounded, the voltage at the corresponding pin in column 2 shall be specified in column 3 of Table IX for conformance with 3.4.3.1.

TABLE IX

1 Pin	2 Pin	3 Voltage More Positive Than:
19B	20B and 15B	-0.6
20A	20B and 15B	-0.6
17A	20B and 15B	-0.6
19A	20B	-0.6
18A	20B	-0.6
18B	15B	-0.6
17B	15B	-0.6

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4.5.2.3.2 Reverse leakage.- Remove all power supplies and test leads from prior tests. Ground pin 58B. Apply a minus 6 volt supply to the indicated pins in column 1, and as minus 6 volts is applied to each pin in column 1, the voltage at the corresponding pin specified in column 2 shall be specified in column 3 of Table X for compliance with 3.4.3.2. At the end of this test remove the ground from pin 58B.

TABLE X

1 Pin	2 Pin	3 Voltage More Positive Than:
19B	20B and 15B	-0.6
20A	20B and 15B	-0.6
17A	20B and 15B	-0.6
58A	20B and 15B	-0.6
19A	20B	-0.6
18A	20B	-0.6
18B	15B	-0.6
17B	15B	-0.6

4.5.2.4 Clamp circuit and spare.

4.5.2.4.1 Forward drop.- Remove all power supplies from the board and connect plus 15 volts to pin 11B. Ground pins 10A and 12B. The voltage measured at pins 15A, 13B, 14B, 14A and 13A shall be less positive than plus 0.8 volt as specified in 3.4.4.1.

4.5.2.4.1.1 Connect a minus 6 volt supply to pin 12A. The voltage measured at pin 13A shall be minus 5.5 ± 0.5 volt dc. Remove minus 6 volts from pin 12A and connect it to pin 11A. The voltage measured at pin 13A shall be minus 5.5 ± 0.5 volt dc.

4.5.2.4.1.2 Connect a 10K ohm, 1/2 watt, ± 1 percent resistor from ground to pins 59A and 60A. Connect the minus 18 volts to pin 59B and 60B. The voltage measured from pin 59B to pin 59A shall be more positive than minus 0.8 volt dc. Remove minus 18 volts from pin 59B and connect it to pin 60B. The voltage measured from pin 60B to pin 60A shall be more positive than minus 0.8 volt. Remove all test leads and supply voltages from the board.

4.5.2.4.2 Reverse leakage.- Ground pin 11B and connect the plus 6 volts to pins 10A and 12B. The voltage measured at those pins listed in column 1 shall be as specified in column 2 of Table XI for compliance with 3.4.4.2.

TABLE XI

1 Pin	2 Voltage More Negative Than:
15A	+ 1.1
13B	+ 0.23
14B	+ 0.23
14A	+ 0.23
13A	+ 0.23

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4.5.2.4.2.1 Remove the plus 6 volts from pins 10A and 12B. Apply the plus 15 volts to pin 11A. The voltage at pin 13A shall be more negative than plus 0.23 volt dc. Remove the plus 15 from pin 11A and apply it to pin 12A. The voltage measured at pin 13A shall be more negative than plus 0.23 volt dc.

4.5.2.4.2.2 Connect to pins 59A and 60A a 10K ohm 1/2 watt, ± 1 percent resistor from ground. Connect plus 15 volts to pin 59B and 60B. The voltage measured at pin 59A shall be more negative than plus 0.50 volt dc. The voltage measured at pin 60A shall be more negative than plus 0.50 volt dc.

4.5.2.4.2.3 Spare resistor check.- Connect plins 65A, 22A, 61A, 64A, 63A, and 62A to minus 18 volts. Ground one end of a 10K ohm, 1/2 watt, ± 1 percent resistor. Connect its other end to each of the pins specified in column 1 of Table XII. When the resistor is connected to a pin in column 1, the voltage at this pin shall be as specified in column 2 of Table XII. At the end of the test remove all test leads and power supplies.

TABLE XII

1 Pin	2 Voltage
22B	-5.7 ± 0.8
61B	-5.7 ± 0.8
65B	-12.3 ± 1.5
64B	-16.3 ± 1.5
63B	-14.7 ± 1.5
62B	-17.3 ± 0.6

4.5.2.5 Anode driver test.

4.5.2.5.1 Reverse leakage.- With all supply voltages removed from the board, ground pin 58B. Apply minus 18 volts sequentially to each of pins 2A, 1B and 1A. The voltage at pin 2B for each application shall be more positive than minus 0.28 volt dc as specified in 3.4.5.1.

4.5.2.5.2 Forward drop.- Remove the ground from pin 58B and apply minus 6 volts. Apply ground sequentially to each of pins 2A, 1B and 1A. The voltage at pin 2B for each application shall be more positive than minus 0.8 volt dc as specified in 3.4.5.2. At the end of this test remove all test leads and supplies from the board.

4.5.2.5.3 False state.- Connect to pin 3B a load as shown in Figure 3. Connect power supplies to indicated pins as specified in Table IV.

4.5.2.5.3.1 False input.- Ground pin 1A and measure the voltage between those pins listed in column 1 and those in column 2 of Table XII.The voltage shall be as listed in column 3 of Table XIII as specified in 3.4.5.4.

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TABLE XIII

1 From Pin	2 To Pin	3 Voltage
TP5	TP4	More negative than +0.2
4B	TP5	More positive than -1.2
4B	TP6	More positive than -1.0
3B	TP6	More negative than +1.0
3B	12B	+66.5 ± 7.5

4.5.2.5.4 True state.- The load and power supplies shall remain connected as specified in 4.5.2.5.3.

4.5.2.5.4.1 True input.- Connect pin 1A to minus 6 volts. Measure the voltage from those pins listed in column 1 to those in column 2 of Table XIV. The voltage shall be as listed in column 3 of Table XIV as specified in 3.4.5.3.

TABLE XIV

1 From Pin	2 To Pin	3 Voltage (volts)
2B	3A	More positive than -1.2
2B	TP4	More positive than -1.0
TP5	4B	More positive than -1.0
56B	4A	+12.5 ± 1.0
4A	3B	More negative than +1.0
3B	TP6	More positive than -1.2
3B	12B	+141 ± 13

4.5.2.5.5 Dynamic test.- With the power supplies and load remaining connected as specified in 4.5.2.5.3 apply to pin 1A an input test signal of 200 cps ± 5 percent varying from 0 ± 1 to minus 6 ± 1 volts as specified in Figure 4. The output waveform at pin 3B shall be as specified in Figure 4. The delay time measured from the time the input is at minus 1.5 volts (going towards minus 6 volts) to the time when the output reaches plus 125 volts (going towards plus 142 volts) shall be less than 100 microseconds. The delay time measured from the time the input is at minus 1.5 volts (going towards zero volts) to the time the output reaches plus 75 volts (going towards plus 68 volts) shall be less than 100 microseconds as specified in 3.4.5.5. At the end of these tests remove all test leads and power supplies from the board.

4.5.2.5.6 Circuit output.- The anode driver output at pin 5B shall meet the requirements of 3.3.3.1 and 3.3.4.1 when tested as outlined in 4.5.2.5.5. At the end of these tests remove all test leads and power supplies from the board.

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4.5.2.6 Cathode driver.

4.5.2.6.1 Output diode.- Remove all power supplies from the board. Ground pin 56B and apply plus 50 volts ± 2 percent to pin 38B. The voltage measured at pins 50A, 48A, 45A, 44A, 37A, 35A, 30A, 28A, 24A, 23A, 23B, 54A and 52A shall be plus 15 ± 14.9 volts dc to determine compliance with 3.4.6.1. Remove all test leads and power supplies from the board after completion of test.

4.5.2.6.2 Selection and input diodes-forward drop.- Connect to pin 40B a minus 18 volt supply voltage. Ground the pins listed in column 1 individually for each test and measure the voltage at the points listed in column 2. The voltage measured at column 2 of Table XV shall be minus 0.6 ± 0.4 volt for compliance with 3.4.6.2.

TABLE XV

1 Ground Pin	2 Measure Voltage at Pin
55B	43B
44B	43B
57B	55A
51B	55A
26A	24B
28B	24B
TP1	24B
26A	26B
26B	25B
49B	49A
49A	25B
41A	26B
26B	25A
49B	TP1
57A	TP1
TP1	25A
28B	29B
29B	29A
34A	32A
40A	32A
32A	29A
43A	29B
29B	31A
34A	33A
33A	31A
29B	36A
40A	49A
49A	36A
29B	38A
TP1	38A
43A	45B
41A	45B
45B	TP3

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TABLE XV (Cond)

1 Ground Pin	2 Measure Voltage at Pin
32A	TP3
45B	46A
57A	33A
33A	46A
45B	TP2
49A	TP2
45B	51A
TP1	51A

4.5.2.6.3 True state.- With the power supplies connected as specified in Table IV, apply minus 6 volts to pins 55B 57B, 26B, 29B, 45B, and 49B. To determine compliance with 3.4.6.3, the voltage measured from pin 56A to the following pins shall be more negative than plus 1.2 volts; pins 43B 55A, 25B, 26B, 25A, 29A, 31A, 36A, 38A, 46A, 51A, TP2 and TP3. The voltage measured from the pins listed in column 1 to those pins listed in column 2 of Table XVI shall be more negative than plus 1.0 volt dc. At the end of this test remove the minus 6 volts from pins 55B, 57B, 26B, 29B. 45B, and 49B.

TABLE XVI

1 From Pin	2 To Pin
50A	51A
48A	TP2
45A	46A
44A	TP3
37A	38A
35A	36A
30A	31A
28A	29A
24A	25A
23A	25B
54A	55A
52A	43B
TP7	24B

4.5.2.6.4 False state.- The power supplies shall remain connected as specified in Table I. Ground pins 55B, 57B, 26B, 29B, 45B and 49B. To determine compliance with 3.4.6.4 the voltage measured from pin 38B to pins 50A, 48A, 45A, 44A, 37A, 35A, 30A, 28A, 24A, 23A, 23B, 54A and 52A shall be more positive than minus 1.0 volt dc. At the end of this test remove the ground from pins 55B 57B, 26B, 29B, 45B and 49B.

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4.5.2.6.5 Dynamic test loaded.- Connect load as shown in Figure 5 to pins 50A, 48A, 45A, 44A, 37A, 35A, 30A, 28A, 24A, 23A, 54A, and 52A, and TP7. The power supplies shall remain connected as specified in Table IV.

4.5.2.6.5.1 Input test signal.- Connect the input test signal as specified in Figure 6 in sequence to those pins listed in column 1 of Table XVII. The output waveform at the pins listed in column 2 of Table XVII shall be as specified in Figure 6.

TABLE XVII

1 Input Signal to Pin	2 Output Waveform at Pin
45B	50A
45B	48A
45B	45A
45B	44A
29B	37A
29B	35A
29B	30A
29B	28A
26B	24A
26B	23A
57B	54A
55B	52A
49B	TP7

4.5.2.6.5.1.1 Delay time.- The delay time measured in Figure 6 shall be as follows; the delay time measuring from the time the input is at minus 3 volts (going towards minus 6 volts) to the time the output reaches plus 40 volts, shall be less than 5 microseconds. The delay time measured from the time the input is at minus 3 volts (going towards zero volts) to the time the output reaches plus 60 volts (going towards plus 78 volts) shall be less than 5 microseconds as specified in 3.4.6.5.

4.5.2.6.6 Circuit output.- The cathode drivers outputs as specified in 4.5.2.6.5.1 shall meet the requirements of 3.3.3.1 and 3.3.4.1 when tested as outlined in 4.5.2.6.5.1.1.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Master Packaging Data Sheet M14002.

6. NOTES

6.1 Intended use.- The Network B circuits consists of the anode and cathode drivers, cable driver and the input inverter. The use of each circuit in the Field Artillery Digital Automatic Computer (FADAC) is as follows:

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(a) The anode and cathode drivers are used in conjunction with each other to turn on and off Nixie tubes (alpha or numerical indicators tubes) on the display panel.

(b) The cable drivers are used in pairs to drive clock rate signals through cables to circuits external to the computer.

(c) The input inverters are used to invert signals from the output of input drivers.

(d) The and gates and the clamp circuits are connected to various points in the computer to mechanize the logical processes which the computer performs.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Selection of applicable level of preservation, packaging and packing.

(c) Applicable stock number.

(d) Applicable packaging data sheet (see 5.1).

(e) Block of serial numbers to identify each network board.

Custodian:

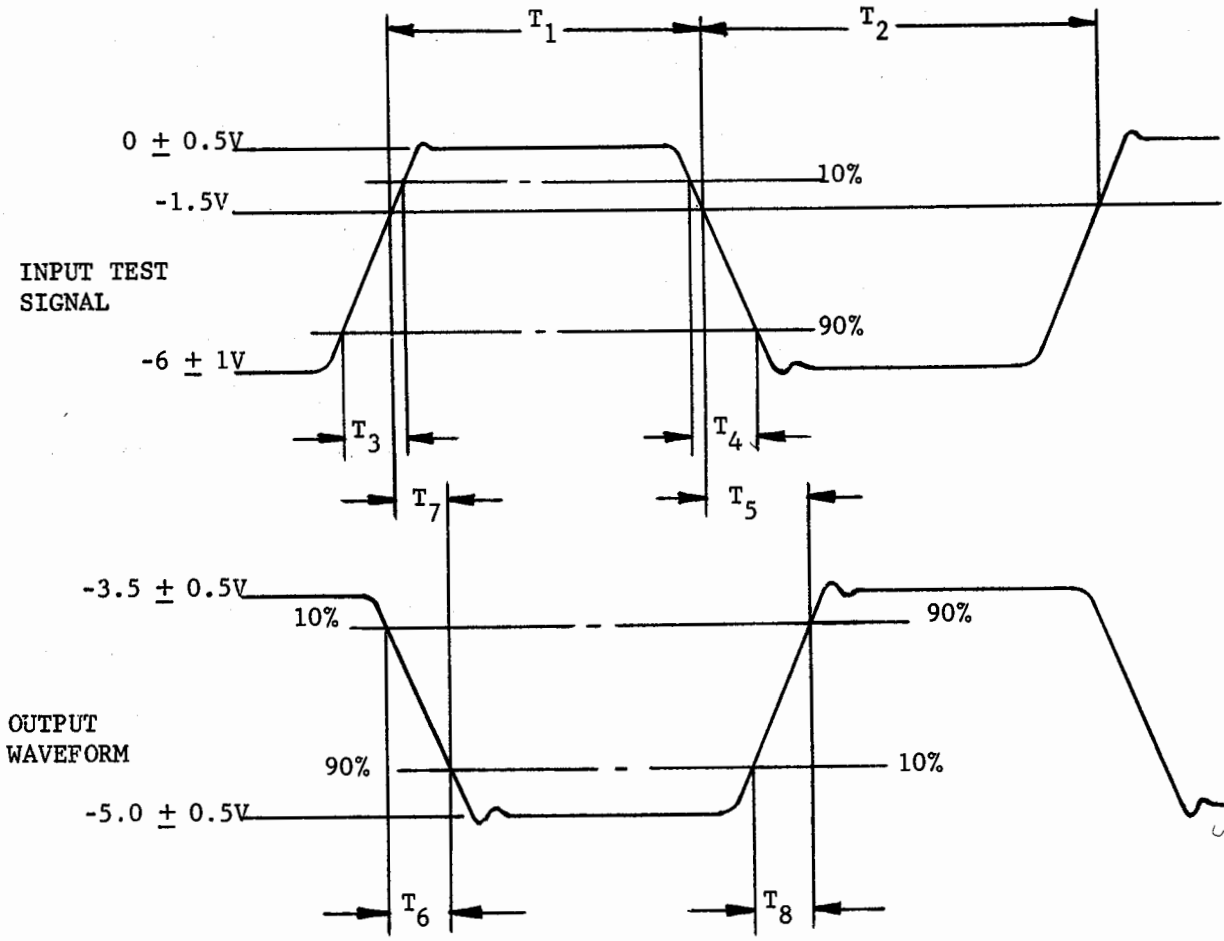
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Preparing activity:

Army - MU

Project No. 1220-A204Y

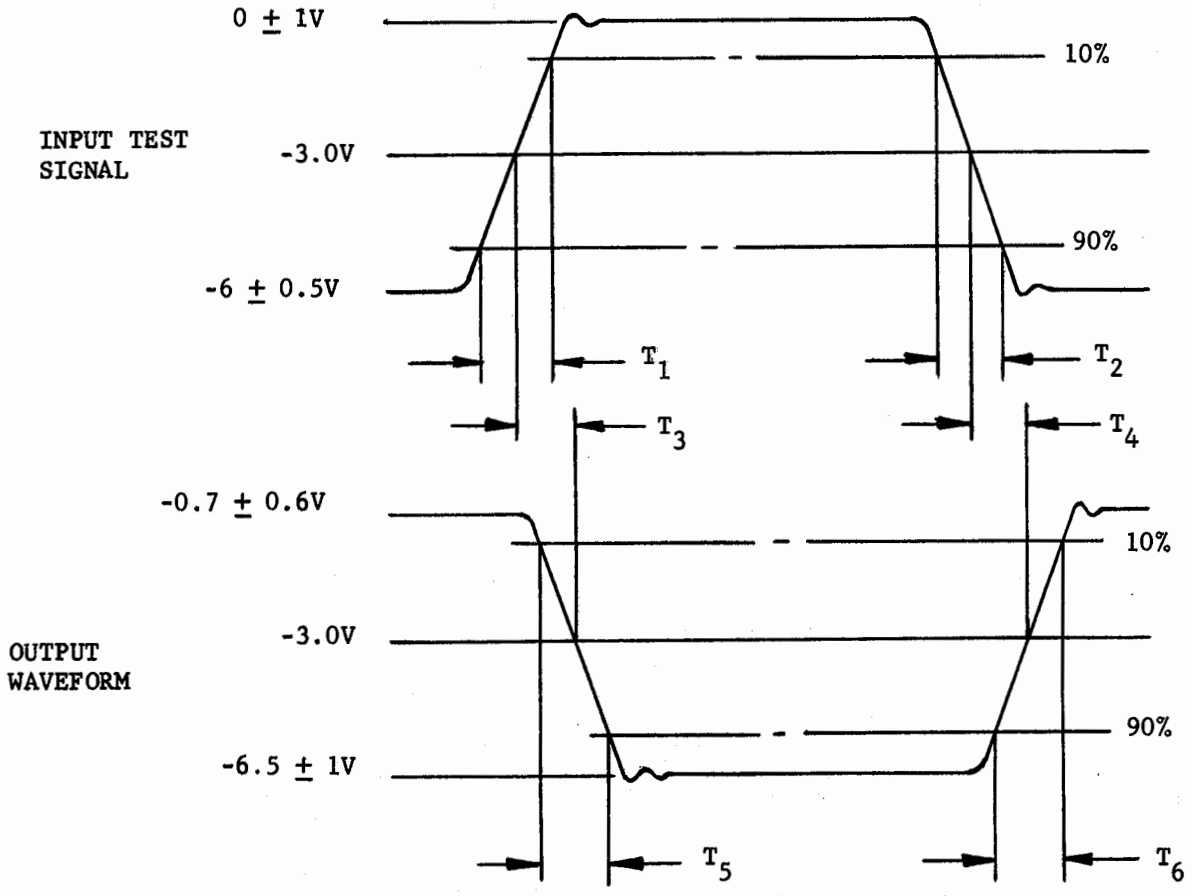
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T_1 = 0.6 ± 0.2 us
T_2 = 1.8 ± 0.2 us
$T_3 = T_4 \leq$ 0.2 us
$T_5 = T_7 \leq$ 0.15 us
$T_6 = T_8 \leq$ 0.15 us

FIGURE 1 - CABLE DRIVER WAVEFORMS

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$T_1 = T_2 \leq 0.5$ us
$T_5 = T_6 \leq 0.5$ us
$T_3 = T_4 \leq 0.5$ us

FIGURE 2 - INPUT INVERTER WAVEFORM

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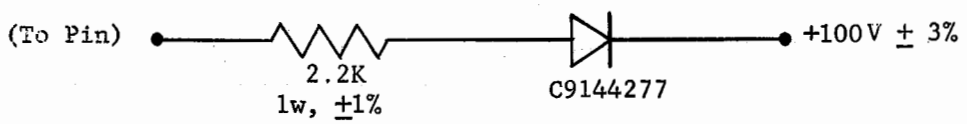
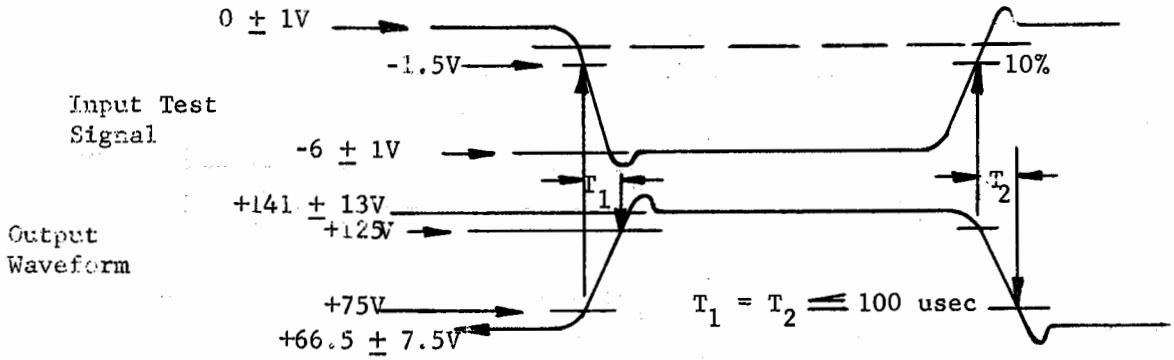


FIGURE 3 — ANODE LOAD



The 10 to 90 percent rise and fall time of the input shall be less than 1 microsecond and of the output shall be less than 50 microseconds.

FIGURE 4 - WAVEFORM ANODE (DRIVER).

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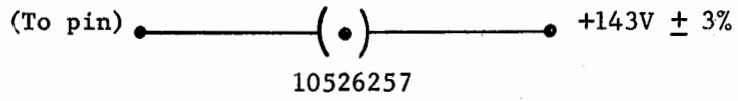
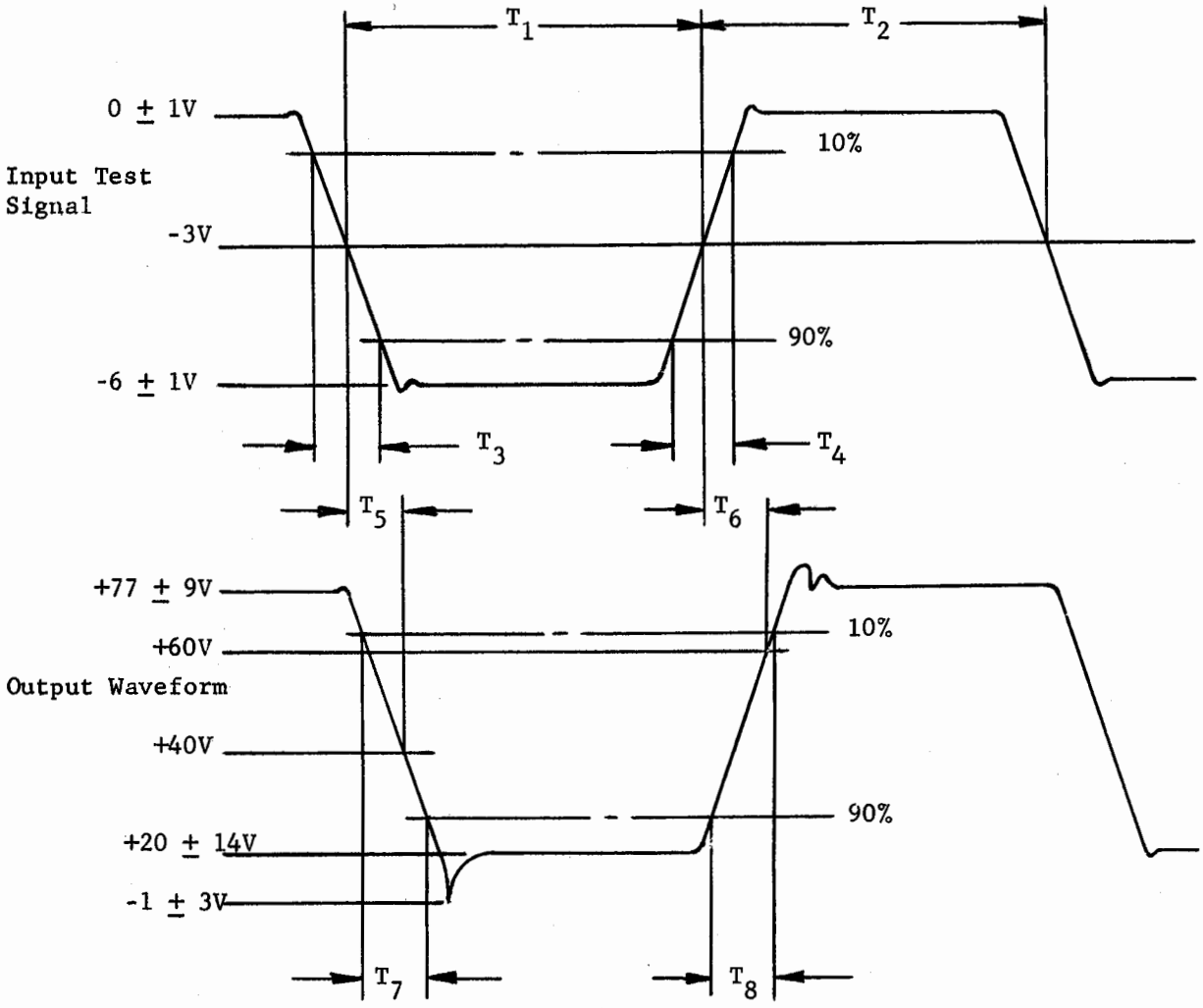


FIGURE 5-CATHODE DRIVER LOAD



$T_1 = T_2 = 50 \pm 1$ us
$T_3 = T_4 \leq 0.5$ us
$T_5 = T_6 \leq 5$ us
$T_7 = T_8 \leq 5.0$ us

FIGURE 6 - CATHODE DRIVER WAVEFORM

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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